LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Force and Laws of Motion

Chapter / Topic: Force and Laws of Motion (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Convert all units to SI before substituting.

SECTION A

6 marks

Choose the correct option.

Q1. The SI unit of force is the: **[1]**

(a) joule (b) watt
(c) newton (d) pascal

Q2. The mass of an object is a measure of its: **[1]**

(a) weight (b) speed
(c) volume (d) inertia

Q3. The working of a rocket is based on: **[1]**

(a) Newton's third law (b) Newton's first law
(c) Newton's second law (d) the law of gravitation

Q4. A force of 10 N acts on a mass of 2 kg. The acceleration produced is: **[1]**

(a) 0.2 m/s^2 (b) 5 m/s^2
(c) 12 m/s^2 (d) 20 m/s^2

Q5. When a stationary bus starts suddenly, the standing passengers tend to fall: **[1]**

(a) forward (b) sideways
(c) backward (d) they don't move at all

Q6. **Assertion (A):** Action and reaction forces do not cancel each other.
Reason (R): Action and reaction act on two different bodies. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why is it easier to push an empty box than a box full of books? **[2]**

OR

Why does dust come out of a carpet when it is beaten with a stick?

Q8. Find the momentum of (a) a 5 kg ball moving at 2 m/s, and (b) a 50 g bullet moving at 200 m/s. Which has more momentum? **[2]**

SECTION C

6 marks

Q9. A car of mass 1000 kg moving at 36 km/h is brought to rest in 5 s. **[3]**

(i) Find its initial momentum. **[1]**

(ii) Find the force needed to stop it. **[1]**

(iii) Why should passengers wear seat belts? **[1]**

Q10. State Newton's second law of motion and use it to show that F = ma. **[3]**

OR

Explain why a gun recoils when a bullet is fired, using the conservation of momentum.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) State the law of conservation of momentum, and prove it using Newton's third law for a collision of two balls. **[3]**

(ii) A 10 g bullet moving at 300 m/s hits a 2 kg wooden block at rest and gets embedded in it. Find their common velocity. **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. In a crash, a car stops almost instantly, but the passengers keep moving forward. Seat belts and airbags protect them. An airbag inflates in front of the passenger and makes the passenger take longer to come to rest. **[4]**

(i) Which law of motion explains why passengers move forward in a crash? **[1]**

(ii) How does an airbag reduce injury? **[1]**

(iii) A 60 kg passenger moving at 20 m/s is stopped (a) by an airbag in 0.5 s, (b) by hitting the dashboard in 0.05 s. Find the force in each case. **[2]**

~ All the best! ~